LAB 9: Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES

- To understand the basic concepts of wireless networking, including SSID, channels, and security types.
- To configure wireless routers and access points in Cisco Packet Tracer.
- To connect multiple devices to a wireless network and verify connectivity.
- To analyze network traffic using Packet Tracer’s simulation mode and troubleshoot issues.

THEORY

Wireless Networks

Wireless networks allow devices to connect and communicate without using physical cables, using radio waves instead. They are widely used in homes, offices, and public spaces due to their flexibility and mobility. Common types include Wireless LAN (WLAN), which covers small areas like homes or offices, and Wireless WAN (WWAN), which covers larger areas using cellular or satellite communication.

Key Wireless Concepts:

Every wireless network has an SSID (Service Set Identifier), which is the network name used by devices to identify and connect. Wireless networks operate on different frequency bands, commonly 2.4 GHz for longer range and 5 GHz for higher speed. Channels within these bands help avoid interference, and security protocols like WEP, WPA, and WPA2 protect data transmitted over the network.

Wireless Devices

A wireless network typically includes access points (APs) and wireless routers. Access points connect wireless devices to the wired network, while wireless routers combine routing and AP functions. Devices like laptops, smartphones, and PCs with wireless adapters act as wireless clients that communicate through these devices.

Packet Analysis

Network communication occurs through packets, which are small units of data containing source and destination information. Cisco Packet Tracer's simulation mode allows users to visualize these packets, see how data flows across the network, and identify potential issues. Packet analysis is useful for troubleshooting problems like IP misconfigurations, authentication failures, or signal interference.

Wireless networks provide mobility, easy device addition, and reduced cabling costs. However, they have limited range, can face interference from other devices, and are vulnerable to security threats if not properly configured.

NETWORK TOPOLOGY

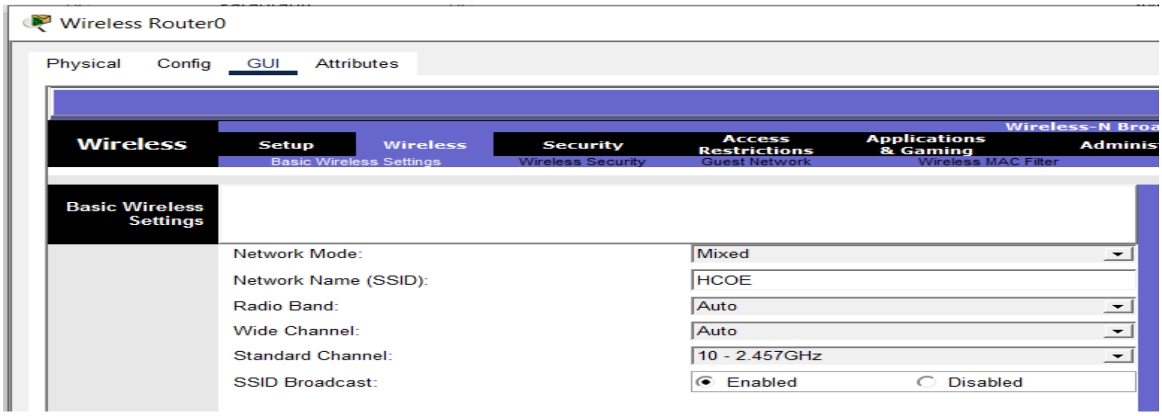
Procedure: Wireless Router Configuration

Step 1: Basic Setup

- Click on the Wireless Router.
- Go to GUI → Setup.
- Configure the following:
  - IP Address: 192.168.0.1
  - Subnet Mask: 255.255.255.0
  - DHCP Server: Enabled

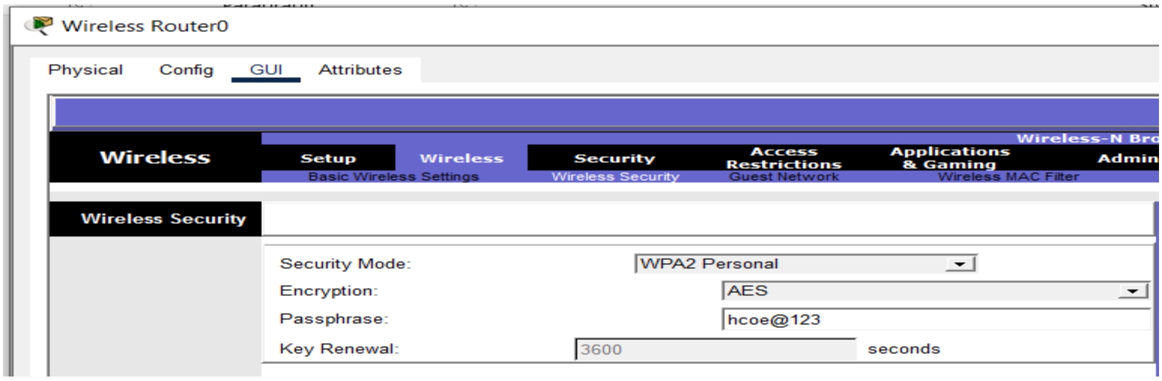
Step 2: Wireless Configuration

- Go to Wireless → Basic Wireless Settings.
- Configure the following parameters:
  - Network Mode: Mixed
  - Network Name (SSID): HCOE
  - Channel: Auto
- Click **Save Settings**



Step 3: Wireless Security

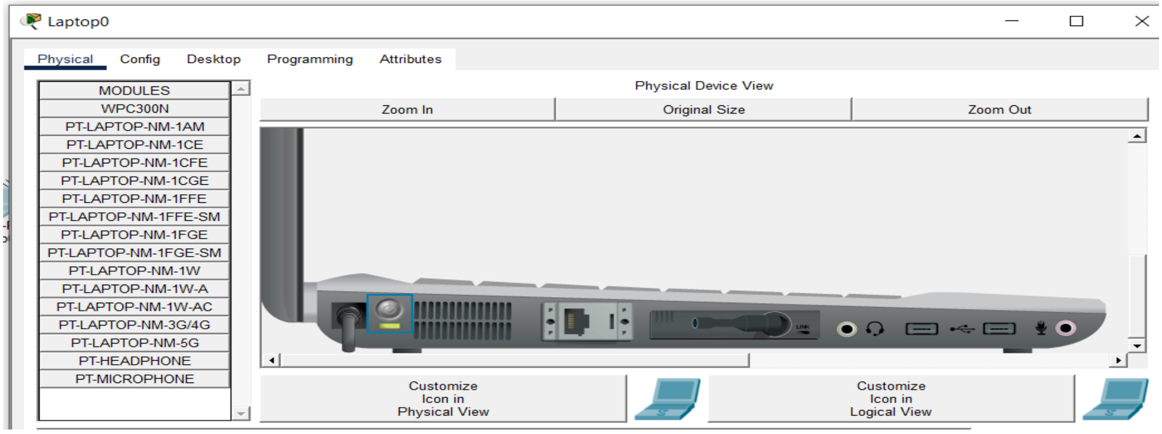
- Go to Wireless → Wireless Security.
- Configure the following:
  - Security Mode: WPA2-Personal
  - Encryption: AES
  - Passphrase: hcoe@123
- Click **Save Settings**



Procedure: Laptop Configuration

Step 1: Check Physical Wireless Adapter (If Needed)

- Click on the Laptop
- Go to the Physical tab.
- Verify if the Wireless Network Adapter is installed (WPC300N Wireless Card).
- If not installed, drag and drop a Wireless Network Adapter into the laptop.
- Ensure the adapter is powered on (green light).



Step 2: Accessing Wireless Network

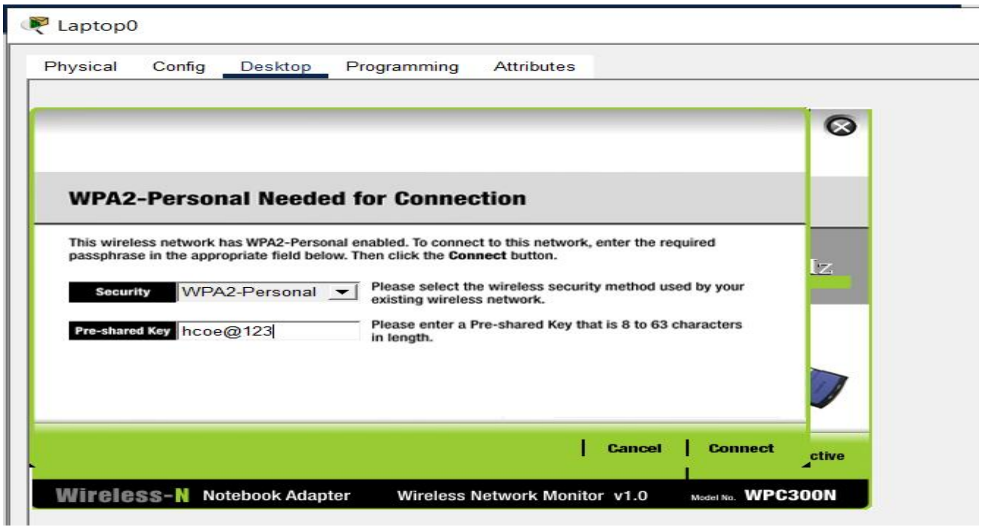
- Go to Desktop → PC Wireless.
- Click on Connect to a Network**.**

Step 3: Selecting the Network

- From the list of available networks, select the SSID: HCOE**.**
- Click **Connect**.

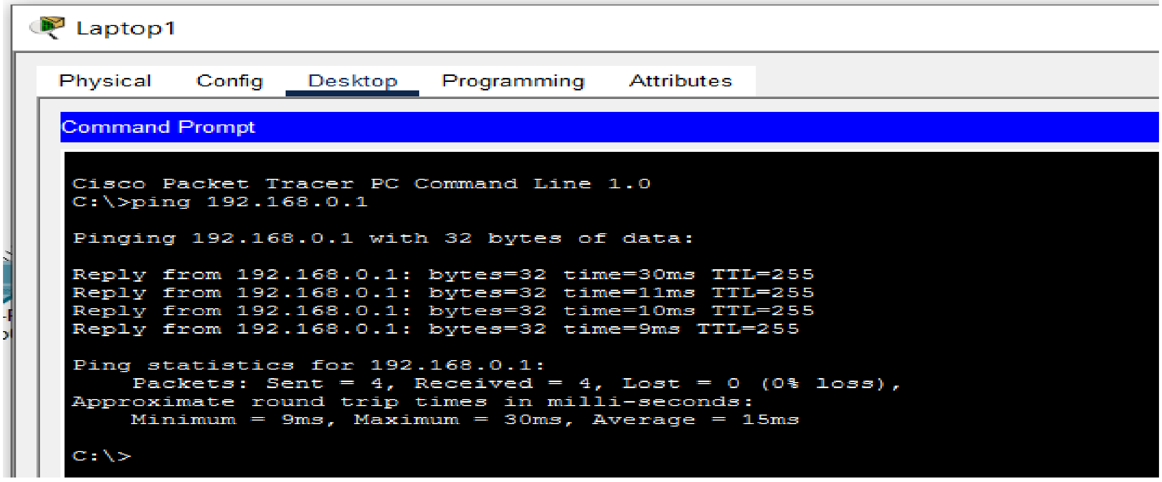
Step 4: Entering Security Details

- Enter the Passphrase: hcoe@123.
- Ensure the connection shows Successful or Connected.



Step 5: Verifying Connectivity

- Go to Desktop → IP Configuration.
- Ensure the laptop has received an IP Address automatically from the router
- Test connectivity using Command Prompt:
    - ping 192.168.0.1 (to check connection to the router)



RESULTS

The wireless network setup in Cisco Packet Tracer was successfully implemented. The WRT300N router was configured with a local IP address of 192.168.0.1, a subnet mask of 255.255.255.0, and an active DHCP server. A wireless network with the SSID "HCOE" was created using WPA2-Personal security, AES encryption, and the passphrase "hcoe@123". Both Laptop0 and Laptop1, equipped with WPC300N wireless adapters, successfully connected to the network and received IP addresses automatically from the DHCP server. Network connectivity was verified through successful ping tests to the default gateway (192.168.0.1) with 0% packet loss, confirming proper communication between all devices.

DISCUSSION

Wireless networking enables devices to communicate without physical cables, offering mobility and reduced setup costs. In this lab, the router acted as both an Access Point and DHCP server, connecting clients and managing the network. Key aspects included network identification through SSID (HCOE), performance using Mixed mode on the 2.4 GHz band, and security with WPA2-Personal encryption. Packet Tracer's Simulation Mode helped visualize packet flow and troubleshoot issues. While wireless networks are flexible, they are limited by signal interference and range compared to wired networks.

CONCLUSION

This lab demonstrated the setup and analysis of a basic WLAN. By configuring the router and connecting laptops, concepts like SSID, DHCP, and WPA2 security were applied effectively. Successful ping tests confirmed reliable communication, showing how wireless networking provides mobility and scalability while emphasizing the need for proper security.